

repres. on 32 bits, SP, $m > 1$.

	C	1	D	E	0	0	0	0
	1	100	0001	1101	1110	0000	0000	0000
s	c = 127 + e (8 bits)				m (23 bits)			

$$c = 10000011_2 = 2^7 + 2^1 + 2^0 = 128 + 2 + 1 = 131$$

$$\Rightarrow 127 + e = 131 \Rightarrow e = 131 - 127 \Rightarrow e = 4$$

$$x = -(\underbrace{1}_{\text{hidden bit}}, 101111_2) \cdot 2^4 = -11011,11_2 =$$

$$= -(2^4 + 2^3 + 2^1 + 2^0 + 2^{-1} + 2^{-2}) =$$

$$= -(16 + 8 + 2 + 1 + \frac{1}{2} + \frac{1}{4}) = -(27 + \frac{3}{4}) = -27,75$$

• Homework presentations

(Ex.1) (2) Check the associativity for the $\downarrow$ (nand) operation

$$(p \downarrow (q \downarrow r)) \equiv (p \downarrow q) \downarrow r$$

$$p \downarrow q \stackrel{\text{def}}{=} \neg(p \vee q)$$

step 1 $u \equiv v$ iff u, v have identical truth table

step 2 Truth table

		u			v		
i	p	q	r	$q \downarrow r$	$p \downarrow (q \downarrow r)$	$p \downarrow q$	$(p \downarrow q) \downarrow r$
i1	F	F	F	T	F	T	F
i2	F	F	T	F	T	T	F
i3	F	T	F	F	T	F	T
i4	F	T	T	F	T	F	F
i5	T	F	F	T	F	F	T
i6	T	F	T	F	F	F	F
i7	T	T	F	F	F	F	T
i8	T	T	T	F	F	F	F

step 3 u and v don't have identical truth tables, so they are not logically equivalent: $u \not\equiv v$, " $\downarrow$ " is not associative

Ex. 2.2 $U_2 = \underbrace{\neg p \vee \neg (q \wedge r)}_u \rightarrow \underbrace{q \wedge \neg p}_v$ } Decide the type of U_2

Step 1 Definitions

1. U is consistent iff $\exists i, i(u) = T$; i -model of U
2. U is inconsistent iff $\forall i, i(u) = F$
3. U is valid (tautology) iff $\forall i, i(u) = T$
4. U is contingent iff $\exists i, i(u) = T$, i -model of U
 $\exists j, j(u) = F$, j -anti-model of U

Step 2 Truth table

i	p	q	r	$\neg p$	$\neg(q \wedge r)$	$\neg p \vee \neg(q \wedge r)$	$q \wedge \neg p$	$U_2 = u \rightarrow v$
i_1	F	F	F	T	T	T	F	F
i_2	F	F	T	T	T	T	F	F
i_3	F	T	F	T	T	T	T	T
i_4	F	T	T	T	F	T	T	T
i_5	T	F	F	F	T	T	F	F
i_6	T	F	T	F	T	T	F	F
i_7	T	T	F	F	T	T	F	F
i_8	T	T	T	F	F	F	F	T

Step 3

U_2 is a contingent formula having 3 models (i_3, i_4, i_8) and 5 anti-models (i_1, i_2, i_5, i_6, i_7)

$i_3: \{p, q, r\} \rightarrow \{T, F\}$

$i_3(p) = T, i_3(q) = T, i_3(r) = F, i_3(U_2) = T \rightarrow$ model

$i_1: \{p, q, r\} \rightarrow \{T, F\}$

$i_1(p) = F, i_1(q) = F, i_1(r) = T, i_1(U_2) = F \rightarrow$ anti-model

Ex. 3.2. Step 1 $\overbrace{p \rightarrow q}^u \models \overbrace{(q \rightarrow r) \rightarrow (p \rightarrow r)}^v$

$u \models v$ (v is a logical consequence of u)

iff $\forall i, i(u) = T$, we have $i(v) = T$ (All the models of u are also models of v)

Step 2 Truth table

i	P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$P \rightarrow R$	V
i1	F	F	F	T	T	T	T
i2	F	F	T	T	T	T	T
i3	F	T	F	T	F	T	T
i4	F	T	T	T	T	T	T
i5	T	F	F	F	T	F	F
i6	T	F	T	F	T	T	T
i7	T	T	F	T	F	F	T
i8	T	T	T	T	F	T	T

Step 3

All the models of u ($i1, i2, i3, i4, i7, i8$) are also models of v ,
so $u \models v$

Ex. 5.2.

DNF $\vee (\wedge l_{ij})_{\text{cubes}}$

$$A \rightarrow B \equiv \neg A \vee B$$

CNF $\wedge (\vee l_{ij})_{\text{clauses}}$

$$U_2 = (p \rightarrow q) \wedge (p \wedge q \rightarrow r) \xrightarrow{3} (p \rightarrow r). \text{ Write DNF, CNF}$$

We apply the normalization alg.

$$U_2 \xrightarrow{\text{replace } 1, 2} (\neg p \vee q) \wedge (\neg(p \wedge q) \vee r) \xrightarrow{3} (\neg p \vee r)$$

$$\xrightarrow{\text{replace } 3} \neg((\neg p \vee q) \wedge (\neg(p \wedge q) \vee r)) \vee (\neg p \vee r)$$

$$\xrightarrow{\text{De Morgan's}} (p \wedge \neg q) \vee (p \wedge q \wedge \neg r) \vee \neg p \vee r = \text{DNF with 4 cubes}$$

$$\begin{aligned} &\xrightarrow{\text{distrib. law}} (\underbrace{p \vee p}_{\equiv T} \vee \underbrace{\neg p \vee \neg p}_{\equiv T} \vee r) \wedge (\underbrace{p \vee q}_{\equiv T} \vee \underbrace{\neg p \vee \neg p}_{\equiv T} \vee r) \wedge (\underbrace{p \vee r}_{\equiv T} \vee \underbrace{\neg p \vee \neg p}_{\equiv T} \vee r) \wedge \\ &\quad \wedge (\underbrace{\neg q \vee p \vee \neg p \vee r}_{\equiv T}) \wedge (\underbrace{\neg q \vee q \vee \neg p \vee r}_{\equiv T}) \wedge (\underbrace{\neg q \vee r \vee \neg p \vee r}_{\equiv T}) = \\ &= \text{CNF with 6 clauses} \equiv T \wedge T \wedge T \wedge T \wedge T \wedge T \equiv T \end{aligned}$$

$$* \quad p \vee \neg p \vee r \equiv T$$

$$p \wedge \neg p \wedge \neg q \equiv F$$

1. u is valid (tautology) if all the clauses of CNF(u) are valid. A clause is valid if it contains a pair of opposite literals.

2. u is inconsistent if all the cubes of DNF(u) are inconsistent. A cube is inconsistent if it contains a pair of opposite literals.

Models:

- i_1 ($i_1 = i_8 = i_4$)
- i_2 ($i_2 = i_5 = i_7$)
- i_3
- i_6

THEORETICAL ASPECTS:

- Normalization algorithm

- 1) replace $\rightarrow, \leftrightarrow$: $p \rightarrow q \equiv \neg p \vee q$
- 2) apply DeMorgan's laws
- 3) apply distributivity

Ex. 8

$$\text{CNF: } (p \vee q) \wedge (\neg v \vee \neg v) \wedge (\neg v) \wedge (\neg v)$$

Clause 1: F

$$p \vee q \quad F \Rightarrow p, q \rightarrow F$$

$$\neg v \rightarrow T, F$$

Ex. 9

Use definition of deduction to prove the following deductions:

$$p \rightarrow \pi, p \vee \pi \rightarrow q, \pi \vdash q$$

THEORETICAL ASPECTS

The sequence $(f_1, \dots, f_m)$ is the deduction of c from the hypothesis (and the axioms)

$$f_1: p \rightarrow \pi$$

$$f_2: p \vee \pi \rightarrow q \equiv (\neg p \rightarrow \pi) \rightarrow q$$

$$f_3: \pi$$

$$f_4: \pi \rightarrow (\neg p \rightarrow \pi) \text{ obtained from } A_1$$

$$f_3, f_4 \vdash_{\text{mp}} \neg p \rightarrow \pi = f_5 \text{ from } A_1 \text{ with } \begin{cases} U = \pi \\ V = \neg p \end{cases}$$

$$f_2, f_5 \vdash_{\text{mp}} q = f_6 \quad \begin{cases} U = (\neg p \rightarrow \pi) \\ V = q \end{cases}$$

• Axioms:

$$A_1: U \rightarrow (V \rightarrow U)$$

$$U = \pi$$

$$V = \neg p$$

MODUS PONENS

$$U, U \rightarrow V \vdash_{\text{mp}} V$$

Ex. 11

Use theorem of deduction & its reverse to prove:

$$\vdash (p \rightarrow q) \rightarrow ((\neg \pi \vee p) \rightarrow (\pi \rightarrow q))$$

• Step 1) APPLY REVERSE OF THEOREM OF DEDUCTION

$$\text{if } \vdash (p \rightarrow q) \rightarrow ((\neg \pi \vee p) \rightarrow (\pi \rightarrow q))$$

$$p \rightarrow q \vdash (\neg \pi \vee p) \rightarrow (\pi \rightarrow q) \text{ then}$$

$$p \rightarrow q, \neg \pi \vee p \vdash \pi \rightarrow q \text{ then}$$

$$p \rightarrow q, \neg \pi \vee p, \pi \vdash q$$

• Step 2) PROVE DEDUCTION

$$f_1: \pi$$

$$f_2: \neg \pi \vee p \equiv \pi \rightarrow p$$

$$f_3: p \rightarrow q$$

$$f_1, f_2: \vdash_{\text{mp}} p$$

$$f_4: p$$

$$f_3, f_4: \vdash_{\text{mp}} q = f_5$$

• Step 3) USE THEOREM OF DEDUCTION

$$\text{if } U_1, U_2, \dots, U_m \vdash V$$

$$U_1, U_2, \dots, U_{m-1} \vdash U_m \rightarrow V$$

$$\text{if } \pi, \neg \pi \vee p, p \rightarrow q \vdash q$$

$$\text{then } \neg \pi \vee p, p \rightarrow q \vdash \pi \rightarrow q$$

$$\text{then } p \rightarrow q \vdash (\neg \pi \vee p) \rightarrow (\pi \rightarrow q)$$

$$\text{then } \vdash U_2$$

* mp = MODUS PONENS

Ex 12

H_1 : it is not sunny and cold

H_2 : We will go swimming only if sunny.

H_3 : if we don't go swimming, we will take canoe trip.

H_4 : if we take canoe trip, home by sunset.

C : Home by sunset

M - it is sunny

T - it is cold

N - We will go swimming

P - We will take a canoe trip

Q - We will be home by sunset

H_1 : $\neg M \wedge T$

H_2 : $N \rightarrow M \equiv \neg N \vee M$

H_3 : $\neg N \rightarrow P$

H_4 : $P \rightarrow Q$

C : Q

$f_1 = H_1$

$f_2 = H_2$

$f_3 = H_3$

$f_4 = H_4$

$f_2 \vdash_{mt} \neg N \rightarrow \neg M = f_5$

$f_1, f_5 \vdash_{mt} \neg N = f_6$
 $f_1 \vdash_{simplified} \neg M$

$f_6, f_3 \vdash_{mp} P = f_7$

$f_7, f_4 \vdash_{mp} Q = C$

MODUS TOLLENS: $\neg$

$U \rightarrow V \vdash \neg V \rightarrow \neg U$

SIMPLIFICATION

$U \wedge V \vdash U$

$U \wedge V \vdash V$

• PREDICATE LOGIC

Ex 1 Transform the following sentences from natural language into predicate formulas

- VARIABLES
- CONSTANTS
- FUNCTIONS
- PREDICATES

1.2 in a plane there are lines parallel to a constant line d and there are lines perpendicular to d

$$\neg (\exists x)(\exists y)(\text{parallel}(x,d) \wedge \text{perpendicular}(y,d)) \rightarrow (\exists x)(\text{parallel}(x,d)) \wedge (\exists y)(\text{perpendicular}(y,d))$$

• VARIABLES : x, y

CONSTANTS : d

FUNCTIONS : $-$

$$\left(\begin{array}{l} \text{sum}(x,y) = x+y \quad \text{sum} : \mathbb{R} \rightarrow \mathbb{R} \\ f : \mathbb{N} \rightarrow \mathbb{N}, f(x) = x^2 \end{array} \right)$$

PREDICATES :

parallel : $D \times D \rightarrow \{T, F\}$

parallel(x, y) = " $x \parallel y$ "

perpendicular $D \times D \rightarrow \{T, F\}$

perpendicular(x, y) = " $x \perp y$ "

REFLEXIVITY :

$$\begin{array}{c} \parallel \\ x \parallel x \quad \checkmark \end{array}$$

$$\begin{array}{c} \perp \\ x \perp x \end{array}$$

SYMMETRY :

$$x \parallel y \Rightarrow y \parallel x$$

$$x \perp y \Rightarrow y \perp x$$

TRANSITIVITY :

$$\begin{array}{l} x \parallel y \wedge y \parallel z \\ \Rightarrow x \parallel z \end{array}$$

$$x \perp y \wedge y \perp z \Rightarrow x \perp z$$

REFLEXIVITY : $(\forall x) \text{parallel}(x, x)$

SYMMETRY : $(\forall x)(\forall y)(\text{parallel}(x, y) \rightarrow \text{parallel}(y, x))$

TRANSITIVITY : $(\forall x)(\forall y)(\forall z)(\text{parallel}(x, y) \wedge \text{parallel}(y, z) \rightarrow \text{parallel}(x, z))$

- 1.8. The sum of two even numbers is an even number and their product is divisible by 4.

$$D = \mathbb{N}, \quad 4 \in D, \quad 4 - \text{const.}$$

Predicate symbols: $eq: D \times D \rightarrow \{T, F\}$, $eq(x, y) = T$ if $x = y$

$div: D \times D \rightarrow \{T, F\}$, $div(x, y) = T$ if $x : y$

Function symbols:

$sum: D \times D \rightarrow D$, $sum(x, y) = "x + y"$

$prod: D \times D \rightarrow D$, $prod(x, y) = "x \cdot y"$

$$(\forall x)(\forall y)(div(x, 2) \wedge div(y, 2) \rightarrow div(sum(x, y), 2) \wedge div(prod(x, y), 4))$$

Properties of div :

- reflexive: $(\forall x) div(x, x)$

- transitive: $(\forall x)(\forall y)(\forall z)(div(x, y) \wedge div(y, z) \rightarrow div(x, z))$

Properties of $sum, prod$:

* sum :

- commutativity: $(\forall x)(\forall y)(eq(sum(x, y), sum(y, x)))$

- associativity: $(\forall x)(\forall y)(\forall z)[eq(sum(x, sum(y, z)), sum(sum(x, y), z))]$

- distributivity:

$(\forall x)(\forall y)(\forall z)[eq(prod(x, sum(y, z)), sum(prod(x, y), prod(x, z)))]$

- 2.12. Every animal either likes to eat all plants or all animals much smaller than itself that like to eat some plants.

A - animals ; P - plants

• Predicate symbols:

$sm: A \times A \rightarrow \{T, F\}$, $sm(x, y) = T$ if x is much smaller than y

$et: A \times (A \cup P) \rightarrow \{T, F\}$, $et(x, y) = T$ if x likes to eat y

$$(\forall x)_{x \in A} \left[(\forall y)_{y \in P} et(x, y) \vee (\forall z)_{z \in A} (sm(z, x) \wedge (\exists t)_{t \in P} et(z, t) \rightarrow et(x, z)) \right]$$

• Propri:

- transitivity for sm :

$$(\forall x)(\forall y)(\forall z)[sm(x, y) \wedge sm(y, z) \rightarrow sm(x, z)]$$

Ex. 7.2. $U_2 = (\neg p \vee q) \wedge \neg(\neg q \rightarrow \neg p)$

We apply the normalization algorithm.

replace $\rightarrow$
 $\equiv (\neg p \vee q) \wedge \neg(q \vee \neg p)$

De Morgan's
 $\equiv (\neg p \vee q) \wedge \neg q \wedge p = \text{CNF with 3 clauses}$

distrib. laws
 $\equiv (\underbrace{\neg p \vee \neg q}_{\equiv F} \wedge p) \vee (\underbrace{q \wedge \neg q}_{\equiv F} \wedge p) = \text{DNF with 2 cubes}$

$\equiv F \vee F = F \Rightarrow U_2\text{-inconsistent}$

Homework (in 2 weeks) $\rightarrow$ ex. 5.2

SEMINAR 6

Ex. 8 Write all the anti-models of U_1 using the appropriate THEORETICAL NF.

$U_2 = (q \vee \pi \rightarrow p) \rightarrow (p \rightarrow \pi) \wedge q$

Ex. 6 Using the appropriate NF, write all models of U_2

$U_2 = \neg(\neg p \vee q) \vee \pi \rightarrow \neg p \wedge \neg(q \wedge \pi)$

$U_2 = \neg(\neg(\neg p \vee q) \vee \pi) \vee \neg p \wedge \neg(q \wedge \pi)$

$U_2 = ((\neg p \vee q) \wedge \neg \pi) \vee (\neg p \wedge (q \vee \neg \pi))$

$U_2 = (\neg p \wedge \neg \pi) \vee (q \wedge \neg \pi) \vee (\neg p \wedge q) \vee (\neg p \wedge \pi)$

Disjunctive form with 4 cubes

• DNF

Cube 1: $\neg p \wedge \neg \pi$
 $\begin{array}{cc} \neg p & \neg \pi \\ \downarrow & \downarrow \\ p=F & \pi=F \end{array}$

$i: \{p, q, \pi\} \rightarrow \{T, F\}$
 $i_1(p) = F$
 $i_1(q) = T$
 $i_1(\pi) = F$

Cube 2: $q \wedge \neg \pi$
 $\begin{array}{cc} i_2(p) = T & \neg \pi \\ i_2(q) = T & \neg \pi \\ i_2(\pi) = F & \neg \pi \end{array}$

Cube 3: $\neg p \wedge q$
 $\begin{array}{cc} i_3(p) = F & q \\ i_3(q) = F & q \\ i_3(\pi) = F & q \end{array}$

Cube 4: $\neg p \wedge \pi$
 $\begin{array}{cc} i_4(p) = F & \pi \\ i_4(q) = F & \pi \\ i_4(\pi) = F & \pi \end{array}$

4.2 Using the given interpretations, evaluate the following formulas:

$$U = (\exists x)(P(x) \wedge Q(x)) \rightarrow (\exists x)P(x) \vee Q(12)$$

interpretation: $i = \langle D, m \rangle$, where $D = \mathbb{N}$

$$m(P) : \mathbb{N} \rightarrow \{T, F\}, m(P)(x) = "x : 5"$$

$$m(Q) : \mathbb{N} \rightarrow \{T, F\}, m(Q)(x) = "x : 7"$$

$$\begin{aligned} v^i(U) &= v^i((\exists x)(P(x) \wedge Q(x)) \rightarrow (\exists x)P(x) \vee Q(12)) \\ &= v^i((\exists x)(P(x) \wedge Q(x))) \rightarrow v^i((\exists x)P(x) \vee Q(12)) \\ &= (\exists x)_{x \in \mathbb{N}} ((x : 5) \wedge (x : 7)) \rightarrow (\exists x)_{x \in \mathbb{N}} (x : 5) \vee (12 : 7) \\ &= (T \wedge T) \rightarrow T \vee F \\ &= T \rightarrow T = T \end{aligned}$$

$\Rightarrow i$ is a model of U

6.2 $U = (\exists x)(P(x) \rightarrow Q(x)) \rightarrow ((\exists x)P(x) \rightarrow (\exists x)Q(x))$

• Find an anti-model:

$i = \langle D, m \rangle$, where $D = \{2m \mid m \in \mathbb{N}\} \cup \{3\}$

$$m(P) : D \rightarrow \{T, F\}, m(P)(x) = "x \text{ is odd}"$$

$$m(Q) : D \rightarrow \{T, F\}, m(Q)(x) = T \text{ if } x = 5$$

$$\begin{aligned} v^i(U) &= v^i((\exists x)(P(x) \rightarrow Q(x)) \rightarrow ((\exists x)P(x) \rightarrow (\exists x)Q(x))) \\ &= v^i((\exists x)(P(x) \rightarrow Q(x))) \rightarrow (v^i((\exists x)P(x)) \rightarrow v^i((\exists x)Q(x))) \\ &= (\exists x)_{x \in D} (P(x) \rightarrow Q(x)) \rightarrow ((\exists x)_{x \in D} P(x) \rightarrow (\exists x)_{x \in D} Q(x)) \\ &= (\exists x)_{x \in D} (\text{not } "x \text{ is odd}" \vee x = 5) \rightarrow ((\exists x)_{x \in D} "x \text{ is odd}" \rightarrow (\exists x)_{x \in D} "x = 5") \\ &= (T \vee F) \rightarrow (T \rightarrow F) \\ &= T \rightarrow F = F \end{aligned}$$

$\Rightarrow i$ is an anti-model of U

SEMINAR 9

• First order predicate formulas

$$H_1: (\forall x) (\overset{ch}{\text{child}}(x) \rightarrow \text{loves}(x, \overset{sa}{\text{Santa}}))$$

$$H_2: (\forall x) (\forall y) (\text{loves}(x, \overset{sa}{\text{Santa}}) \wedge \text{reindeer}(y) \rightarrow \text{loves}(x, y))$$

$$H_3: \overset{re}{\text{reindeer}}(\overset{R}{\text{Rudolf}}) \wedge \overset{R}{\text{red-nose}}(\overset{R}{\text{Rudolf}})$$

$$H_4: (\forall z) (\overset{nm}{\text{red-nose}}(z) \rightarrow \text{weird}(z) \vee \overset{cl}{\text{clown}}(z))$$

$$H_5: (\forall s) (\text{reindeer}(s) \rightarrow \neg \text{clown}(s))$$

$$H_6: (\forall t) (\text{weird}(t) \rightarrow \neg \overset{sc}{\text{loves}}(\text{Scrooge}, t))$$

$$C: \neg \text{child}(\text{Scrooge})$$

$$H_3 \vdash_{\text{simple}} re(R) : f_7$$

$$H_3 \vdash_{\text{simple}} nm(R) : f_8$$

$$H_1 \vdash_{\text{univ. inst.}}^{[x \leftarrow Sc]} ch(Sc) \rightarrow \text{loves}(Sc, Sa) : f_9$$

$$H_2 \vdash_{\text{univ. inst.}}^{[x \leftarrow Sc]} (\forall y) (\text{loves}(Sc, Sa) \wedge re(y) \rightarrow \text{loves}(Sc, y)) : f_{10}$$

$$f_{10} \vdash_{\text{univ. inst.}}^{[y \leftarrow R]} \text{loves}(Sc, Sa) \wedge re(R) \rightarrow \text{loves}(Sc, R) : f_{11}$$

$$H_4 \vdash_{\text{univ. inst.}}^{[z \leftarrow R]} nm(R) \rightarrow \text{weird}(R) \vee cl(R) : f_{12}$$

$$H_5 \vdash_{\text{univ. inst.}}^{[s \leftarrow R]} re(R) \rightarrow \neg cl(R) : f_{13}$$

$$H_6 \vdash_{\text{univ. inst.}}^{[t \leftarrow R]} \text{weird}(R) \rightarrow \neg \text{loves}(Sc, R) : f_{14}$$

$$f_8, f_{12} \vdash_{\text{mp}} \text{weird}(R) \vee cl(R) : f_{15}$$

$$f_7, f_{13} \vdash_{\text{mp}} \neg cl(R) : f_{16}$$

$$f_{15}, f_{16} \vdash_{\text{res.}} \text{weird}(R) : f_{17}$$

$$f_{17}, f_{14} \vdash_{\text{mp}} \neg \text{loves}(Sc, R) : f_{18}$$

$$f_{11}, f_{18} \vdash_{\text{mt}} \neg (\text{loves}(Sc, Sa) \wedge re(R)) \equiv \neg \text{loves}(Sc, Sa) \vee \neg re(R) : f_{19}$$

$$f_{19}, f_7 \vdash_{\text{res.}} \neg \text{loves}(Sc, Sa) : f_{20}$$

$$f_9, f_{20} \vdash_{\text{mt}} \neg ch(Sc) : f_{21}$$

The sequence $(H_1, \dots, H_6, f_7, \dots, f_{21} = C)$ is the proof (deduction) of C from the hypothesis.

★ inference rules

$$U, U \rightarrow V \vdash_{mp} V$$

$$\neg V, U \rightarrow V \vdash_{mt} \neg U$$

$$U \rightarrow V, V \rightarrow Z \vdash_{sylog} U \rightarrow Z$$

$$U \wedge V \vdash_{simplif} U$$

$$(\forall x) U(x) \vdash_{univ. inst.} U(t)$$

$$\neg U, U \vee V \vdash_{res} V$$

Exercises - Semantic Tableaux Method

(1.2.) Decide what kind (consistent, inconsistent, valid) of formula is U_2 .

$$U_2 = (p \vee q \rightarrow r) \rightarrow (p \vee r \rightarrow q)$$

★ Decomposition rules

α rules

$$A \wedge B$$

|

$$A$$

|

$$B$$

$$\neg(A \vee B)$$

|

$$\neg A$$

|

$$\neg B$$

$$\neg(A \rightarrow B)$$

|

$$A$$

|

$$\neg B$$

β rules

$$A \vee B$$

/ \

$$A \quad B$$

$$(\neg A \vee B)$$

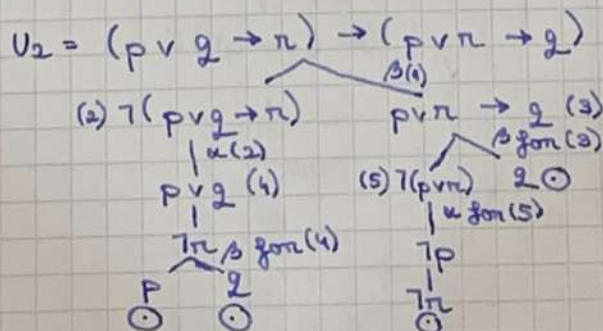
$$A \rightarrow B$$

/ \

$$\neg A \quad B$$

★ Theoretical result

1. A branch is closed if it contains a pair of opposite literals. ($p, \neg p$)
2. U is consistent if it has an open semantic tableaux, at least one open branch.
3. The open branches of U provide the models of U .



The semantic tableaux of U_2 is open with 4 open branches, so U_2 is consistent.

$$\begin{aligned} \text{DNF}(U_2) &= (p \wedge \neg \pi) \vee (q \wedge \pi) \vee (\neg \pi \wedge \neg p) \vee q \\ &\equiv (p \wedge \neg \pi) \vee (\neg \pi \wedge \neg p) \vee q \equiv [\neg \pi \wedge (p \vee \neg p)] \vee q \\ &\equiv \neg \pi \vee q \end{aligned}$$

• Cube $\neg \pi \equiv T$ provides 4 models:

$$i_1, i_2, i_3, i_4 : \{p, q, \pi\} \rightarrow \{T, F\}$$

$$i_1(p) = T \quad i_1(q) = T$$

$$i_2(p) = T \quad i_2(q) = F$$

$$i_3(p) = F \quad i_3(q) = T$$

$$i_4(p) = F \quad i_4(q) = F$$

$$i_{1,2,3,4}(\pi) = F$$

• Cube $q \equiv T$ provides 4 models:

$$i_5, i_6, i_7, i_8 : \{p, q, \pi\} \rightarrow \{T, F\}$$

$$i_5(p) = T \quad i_5(\pi) = T$$

$$i_6(p) = T \quad i_6(\pi) = F \quad i_{5,6,7,8}(q) = T$$

$$i_7(p) = F \quad i_7(\pi) = T$$

$$i_8(p) = F \quad i_8(\pi) = F$$

$$i_1 = i_6, \quad i_3 = i_8$$

U_2 has 6 models:

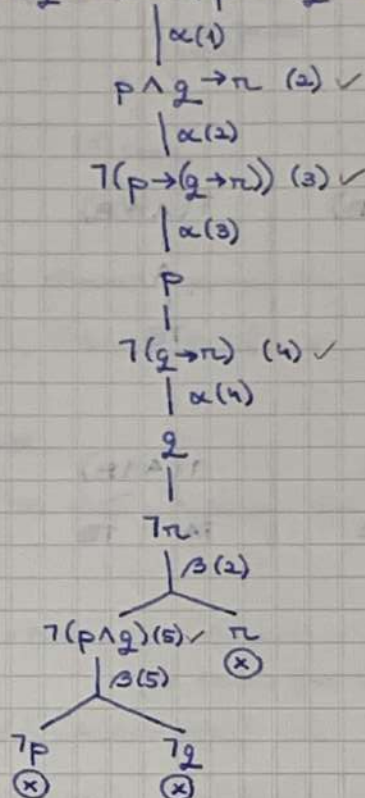
$$i_1(U_2) = i_2(U_2) = i_3(U_2) = i_4(U_2) = i_5(U_2) = i_7(U_2) = T$$

★ Theoretical result:

4. U is inconsistent if it has a closed semantic tableaux, all branches are closed.
5. U is valid ($\models U$) if $\neg U$ has a closed semantic tableaux.

2.2. $U = (p \wedge q \rightarrow r) \rightarrow (p \rightarrow (q \rightarrow r))$ (CHECK VALIDITY)

$$\neg U = \neg[(p \wedge q \rightarrow r) \rightarrow (p \rightarrow (q \rightarrow r))] \quad (1) \checkmark$$



$\neg U$ has a closed sem. tabl. with 3 closed branches $(p, \neg p), (q, \neg q), (r, \neg r)$, so $\neg U$ is inconsistent and U is a tautology.

3.2.

$$\overbrace{p \rightarrow (q \rightarrow r)}^{U_1}, \overbrace{\neg p \vee q}^{U_2} \models \overbrace{(p \rightarrow q) \vee r}^V$$

$$U_1 \wedge U_2 \wedge \neg V \quad (1)$$

$$U_1: p \rightarrow (q \rightarrow r)$$

$$U_2: \neg p \vee q$$

$$\neg V: \neg((p \rightarrow q) \vee r) \quad (4)$$

$$\neg(p \rightarrow q) \quad (5)$$

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$U_1 \wedge U_2 \wedge \neg V$ has a closed semantic tableau with 4 closed branches $(p, \neg p), (q, \neg q), (r, \neg r), (q, \neg q)$, so $U_1, U_2 \models V$

SEMINAR 10

• α rules:

$A \wedge B$	$\neg(A \vee B)$	$\neg(A \rightarrow B)$
A	$\neg A$	A
B	$\neg B$	$\neg B$

• β rules:

$A \vee B$	$A \rightarrow B$	$\neg(A \wedge B)$
/	/	/
A B	$\neg A$ B	$\neg A$ $\neg B$

• δ rule:

$(\exists x) U(x)$
|
 $U(c)$
c - new const.

• γ rule:

$(\forall x) U(x)$
|
 $U(c_1)$ c_1, c_m are all constants on the branch
|
 $U(c_m)$
|
 $(\forall x) U(x)$ copy.

7.2. 'V' - semi distributive cover 'V'

Let $U_1 = (\forall x) A(x) \vee (\forall x) B(x) \rightarrow (\forall x) (A(x) \vee B(x))$

$U_2 = (\forall x) (A(x) \vee B(x)) \rightarrow (\forall x) A(x) \vee (\forall x) B(x)$

$\neg U_1 = \neg((\forall x) A(x) \vee (\forall x) B(x) \rightarrow (\forall x) (A(x) \vee B(x)))$ (1)

| α, γ rules for (1)

$(\forall x) A(x) \vee (\forall x) B(x)$ (2)

$\neg((\forall x) (A(x) \vee B(x))) \equiv (\exists x) (\neg A(x) \wedge \neg B(x))$ (3)

| δ for (3)
a - new constant

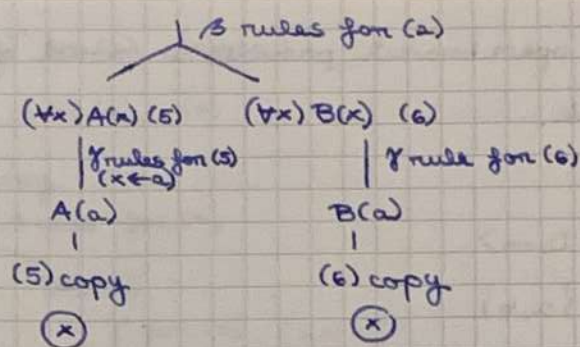
$\neg A(a) \wedge \neg B(a)$ (4)

| α rules for (4)

$\neg A(a)$

|
 $\neg B(a)$

|



$\mathcal{TU}_1$ has a closed semantic tableau with 2 closed branches $(A(a), \neg A(a)); (B(a), \neg B(a))$, so, according to T.s.c. $\models U_1$

★ T.s.c. sem. tabl.

$\models U$ iff $\mathcal{TU}$ has a closed sem. tabl.

$$\mathcal{TU}_2 = \neg [(\forall x)(A(x) \vee B(x)) \rightarrow (\forall x)A(x) \vee (\forall x)B(x)] \quad (1)$$

| α rules for (1)

$$(\forall x)(A(x) \vee B(x)) \quad (2)$$

$$\neg[(\forall x)A(x) \vee (\forall x)B(x)] \quad (3) \quad \checkmark$$

| α rules for (3)

$$\neg(\forall x)A(x) \equiv (\exists x)\neg A(x) \quad (4) \quad \checkmark$$

$$\neg(\forall x)B(x) \equiv (\exists x)\neg B(x) \quad (5) \quad \checkmark$$

| δ rules for (4)
a - new const.

$$\neg A(a) \quad (6)$$

| δ rules for (5)
b - new const.

$$\neg B(b) \quad (7)$$

| γ rules for (2)
a, b - used for inst.

$$A(a) \vee B(a) \quad (8)$$

$$A(b) \vee B(b) \quad (9)$$

(2) copy

| β rules for (8)

$$A(a)$$

 $(\times)$

$$B(a)$$

| β rules for (9)

$$A(b)$$

 $(\circ)$

$$B(b)$$

 $(\times)$

$\mathcal{TU}_2$ has an open semantic tableau, with 2 closed branches $(A(a), \neg A(a)), (B(b), \neg B(b))$, so $\mathcal{TU}_2$ is inconsistent and $\not\models U_2$.

The open branch provides a model of U_2 which is anti model of U_2 .

$$i = \langle D, m \rangle$$

$$D = \{a, b\}$$

$$m(A)(a) = F$$

$$, m(A)(b) = T$$

$$m(B)(a) = T$$

$$, m(B)(b) = F$$

$$v^I(M_2) = T, v^I(U_2) = F$$

Based on the generic model we build a concrete model.

$$i_1 = \langle D_1, m_1 \rangle, D_1 = \{3, 4\}$$

$$m_1(A)(x) = "x \text{ is a perfect square}"$$

$$m_1(B)(x) = "x \text{ is a prime}"$$

5. $H_1, H_2, H_3 \models C$ iff $H_1 \wedge H_2 \wedge H_3 \wedge \neg C$ has a closed sem. tabl.

$$H_1: (\forall x)(H(x) \rightarrow RC(x))$$

$H(x)$ - x is a hummingbird

$$H_2: \neg(\exists x)(L(x) \wedge LOH(x))$$

$RC(x)$ - x is richly colored

$$H_3: (\forall x)(\neg LOH(x) \rightarrow \neg RC(x))$$

$LOH(x)$ - x lives on honey

$$C: (\forall x)(H(x) \rightarrow \neg L(x))$$

$L(x)$ - x is a large bird

$$H_1 \wedge H_2 \wedge H_3 \wedge \neg C \quad (1) \checkmark$$

| α rule (1)

$$H_1: (\forall x)(H(x) \rightarrow RC(x)) \quad (2)$$

|

$$H_2: \neg(\exists x)(L(x) \wedge LOH(x)) \equiv (\forall x)\neg(L(x) \wedge LOH(x)) \quad (3)$$

|

$$H_3: (\forall x)(\neg LOH(x) \rightarrow \neg RC(x)) \quad (4)$$

|

$$\neg C: (\forall x)(H(x) \rightarrow \neg L(x)) \equiv (\exists x)\neg(H(x) \rightarrow \neg L(x)) \quad (5) \checkmark$$

| δ rule for (5)
a-new const.

$$\neg(H(a) \rightarrow \neg L(a)) \quad (6) \checkmark$$

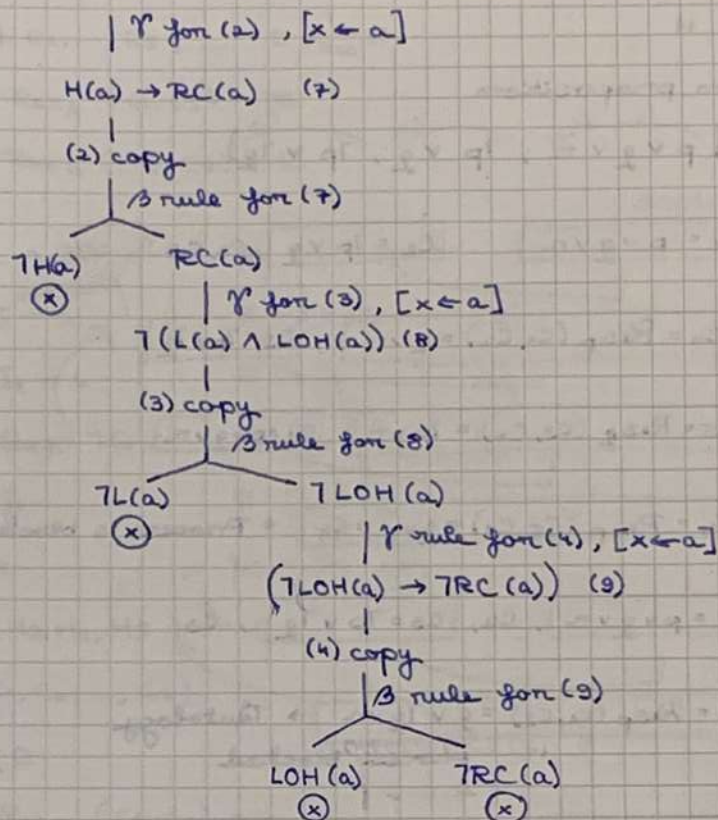
| α rule for (6)

$$H(a)$$

|

$$L(a)$$

|

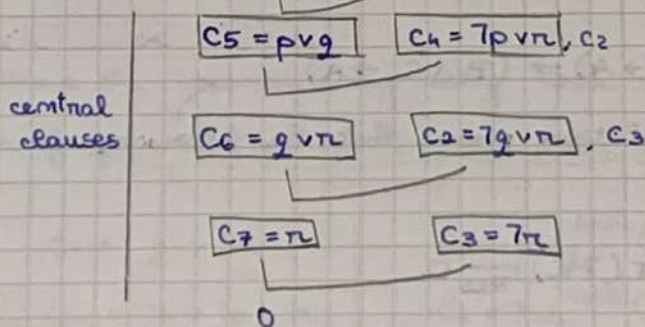


$H_1 \wedge H_2 \wedge H_3 \wedge \neg C$ has a closed semantic tableaux $\Rightarrow C$ is a logical consequence of H_1, H_2, H_3

(1) $S_1 = \{ \underbrace{p \vee q \vee r}_{C_1}, \underbrace{\neg q \vee r}_{C_2}, \underbrace{\neg r}_{C_3}, \underbrace{\neg p \vee r}_{C_4} \}$

(T) S is inconsistent if $S \vdash_{\text{Res}} \square$

top clause: $\boxed{p \vee q \vee r} \quad \boxed{C_3 = \neg r}, C_2, C_1$



side clauses

Resolution

$C_1 = p \vee \neg q$

$C_2 = \neg p \vee r$

$C_3 = \text{Res}_p(C_1, C_2) = \neg q \vee r$

$C_1 = p$

$C_2 = \neg p$

$\text{Res}_p(C_1, C_2) = \square$

Linear resolution

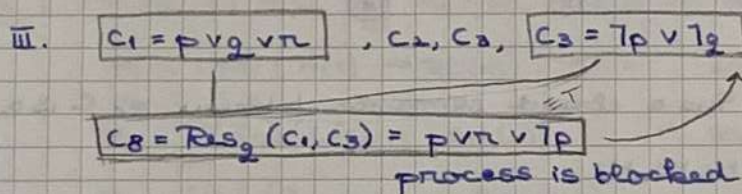
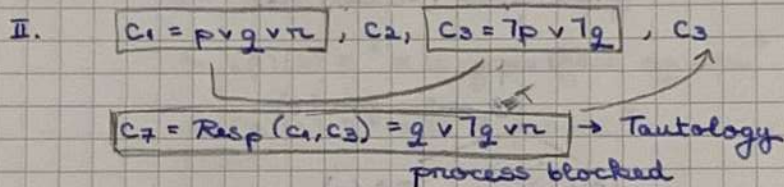
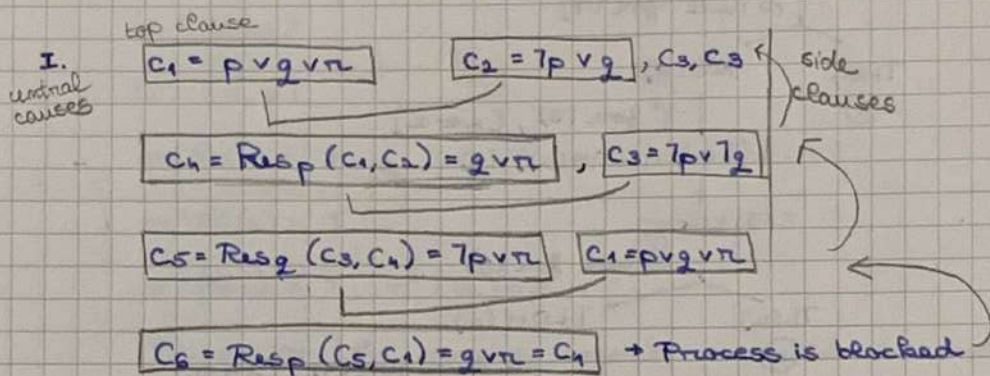
$S_1 \vdash_{\text{Res}}^{\text{lin}} \square, \text{sr}$

S_1 is inconsistent.

SEMINAR 11

Resolution proposition

5.2. $S = \{ \overset{C_1}{p \vee q \vee r}, \overset{C_2}{\neg p \vee q}, \overset{C_3}{\neg p \vee \neg q} \}$



C : After a linear complete search using the backtracking algorithm, without the derivation of the empty clause, we conclude that S is consistent.

1.2. $U_2 = (B \rightarrow A) \wedge (C \rightarrow A) \rightarrow (B \wedge C \rightarrow A)$

Using general resolution prove that the following formula is a theorem.

$(T_2) \vdash U \text{ iff } \text{CNF}(\neg U) \vdash_{\text{Res}} \square$

$\neg U_2 = \neg[(B \rightarrow A) \wedge (C \rightarrow A) \rightarrow (B \wedge C \rightarrow A)] \equiv$

replace $\rightarrow$ by $\neg \vee$

$\equiv \neg[(\neg B \vee A) \wedge (\neg C \vee A) \rightarrow (\neg(B \wedge C) \vee A)]$

replace $\rightarrow$ by $\neg \vee$

$\equiv \neg[\neg[(\neg B \vee A) \wedge (\neg C \vee A)] \vee [(\neg(B \wedge C) \vee A)]]$

$\equiv (\neg B \vee A) \wedge (\neg C \vee A) \wedge (B \wedge C) \wedge \neg A \Rightarrow \text{CNF}$

$\underbrace{\quad}_{C_1} \quad \underbrace{\quad}_{C_2} \quad \underbrace{\quad}_{C_3} \quad \underbrace{\quad}_{C_4} \quad \underbrace{\quad}_{C_5}$

We apply the normalization algorithm

$$S = \{c_1, \dots, c_5\}, \quad S \stackrel{?}{\vdash}_{\text{Res}} \square$$

$$c_6 = \text{Res}_A(c_1, c_5) = \neg B$$

$$c_7 = \text{Res}_B(c_3, c_6) = \square$$

$$(3.2.) \quad H_1 : L \wedge \neg G \rightarrow M \equiv \neg L \vee G \vee M : c_1$$

$$H_2 : J \rightarrow L \equiv \neg J \vee L : c_2$$

$$H_3 : J_t \rightarrow L \equiv \neg J_t \vee L : c_3$$

$$H_4 : G_s \wedge \neg G \equiv \overset{c_4}{G_s} \wedge \neg G : c_5$$

$$H_5 : J_t \equiv J_t : c_6$$

$$C : M, \quad \neg C = \neg M : c_7$$

$$H_1, H_2, H_3, H_4, H_5 \stackrel{?}{\vdash} C$$

$$S = \{c_1, c_2, \dots, c_7\}$$

$$S \stackrel{?}{\vdash}_{\text{Res}} \square$$

$$c_8 = \text{Res}_{J_t}(c_3, c_6) = L$$

$$c_9 = \text{Res}_L(c_1, c_2) = \neg J \vee G \vee M$$

$$c_{10} = \text{Res}_G(c_1, c_5) = \neg L \vee M$$

$$c_{11} = \text{Res}_L(c_8, c_{10}) = M$$

$$c_{12} = \text{Res}_M(c_{11}, c_7) = \square$$

$$\text{CNF}(H_1 \wedge \dots \wedge H_5 \wedge \neg C) \stackrel{?}{\vdash}_{\text{Res}} \square$$

$$\text{so } H_1, \dots, H_5 \vdash C$$

[6.2.] Prove the inconsistency using lock resolution.

$$S = \{ \underbrace{q \vee \neg r}_{c_1}, \underbrace{\neg q \vee \neg p \vee \neg r}_{c_2}, \underbrace{\neg q \vee p \vee \neg r}_{c_3}, \underbrace{r}_{c_4} \}$$

$$a) \quad c_1 = \underbrace{q}_{(1)} \vee \underbrace{\neg r}_{(2)}$$

$$c_2 = \underbrace{\neg q}_{(3)} \vee \underbrace{\neg p}_{(4)} \vee \underbrace{\neg r}_{(5)}$$

$$c_3 = \underbrace{\neg q}_{(6)} \vee \underbrace{p}_{(7)} \vee \underbrace{\neg r}_{(8)}$$

$$c_4 = \underbrace{r}_{(9)}$$

$$c_5 = \text{Res}_q^{\text{lock}}(c_1, c_2) = \underbrace{\neg r}_{(2)} \vee \underbrace{\neg p}_{(4)}$$

$$c_6 = \text{Res}_r^{\text{lock}}(c_4, c_5) = \underbrace{\neg p}_{(4)}$$

$$c_7 = \text{Res}_q^{\text{lock}}(c_1, c_3) = \underbrace{p}_{(7)} \vee \underbrace{\neg r}_{(2)}$$

$$c_8 = \text{Res}_p^{\text{lock}}(c_7, c_6) = \underbrace{\neg r}_{(2)}$$

$$c_9 = \text{Res}_r^{\text{lock}}(c_8, c_4) = \square \Rightarrow S - \text{inconsistent}$$

$$\begin{aligned}
 b) \quad C_1 &= \underset{(2)}{g} \vee \underset{(1)}{7n} \\
 C_2 &= \underset{(3)}{7g} \vee \underset{(1)}{7p} \vee \underset{(3)}{7n} \\
 C_3 &= \underset{(5)}{7g} \vee \underset{(7)}{p} \vee \underset{(6)}{7n} \\
 C_4 &= \underset{(3)}{n}
 \end{aligned}$$

$$S^0 = \{C_1, \dots, C_4\}$$

$$S^1 = \{ \text{Res}_{\text{lock}}^{\text{lock}}(C_i, C_j) \mid C_i \in S^0, C_j \in S^0 \}$$

$$C_5 = \text{Res}_{\pi}^{\text{lock}}(C_1, C_4) = \underset{(2)}{g}$$

$$C_6 = \text{Res}_{\pi}^{\text{lock}}(C_2, C_4) = \underset{(5)}{7g} \vee \underset{(1)}{7p}$$

$$C_7 = \text{Res}_{\pi}^{\text{lock}}(C_3, C_4) = \underset{(6)}{7g} \vee \underset{(7)}{p}$$

$$S^1 = \{C_5, C_6, C_7\}$$

$$S^2 = \{ \text{Res}_{\text{lock}}^{\text{lock}}(C_i, C_j) \mid C_i \in S^1, C_j \in S^0 \cup S^1 \}$$

$$C_8 = \text{Res}_p^{\text{lock}}(C_6, C_7) = 7g$$

$$S^3 = \{ \text{Res}_{\text{lock}}^{\text{lock}}(C_i, C_j) \mid C_i \in S^2, C_j \in S^0 \cup S^1 \cup S^2 \}$$

$$C_9 = \text{Res}_g^{\text{lock}}(C_8, C_5) = \square$$

$$\square \in S^3 \Rightarrow S \text{ is inconsistent}$$

7.2. ★ Check consistency:

$$S_2 = \{ \overbrace{p \vee 7n}^{C_1}, \overbrace{g \vee n}^{C_2}, \overbrace{7p \vee 7n}^{C_3}, \overbrace{7g \vee 7n}^{C_4} \}$$

$$I. \quad S^1 = \{ \text{Res}_p^{\text{lock}}(C_i, C_j) \mid C_i \in S^0, C_j \in S^0 \} = \{C_5, C_6\}$$

$$C_5 = \text{Res}_p^{\text{lock}}(C_1, C_3) = 7n$$

$$C_6 = \text{Res}_n^{\text{lock}}(C_2, C_4) = g \vee 7g \equiv T$$

$$S^2 = \{ \text{Res}_{\text{lock}}^{\text{lock}}(C_i, C_j) \mid C_i \in S^1, C_j \in S^0 \cup S^1 \}$$

$$C_7 = \text{Res}_n^{\text{lock}}(C_5, C_2) = g$$

$$S^3 = \{ \text{Res}_{\text{lock}}^{\text{lock}}(C_i, C_j) \mid C_i \in S^2, C_j \in S^0 \cup S^1 \cup S^2 \}$$

$$S^3 = \emptyset \Rightarrow S_2 - \text{consistent.}$$

After a complete search, using the level saturation strategy, without the derivation of the empty clause, we conclude that S_2 is consistent.

- 1.2. Write the prenex, skolem and clausal forms

$$\begin{aligned}
 U_2 &= (\exists x)(\forall y)\{(\exists z) \neg P(z) \vee (\exists u)[R(x,u) \rightarrow (\forall z) \neg Q(u,z)]\} \\
 &\stackrel{\text{replace}}{\equiv} (\exists x)(\forall y)\{(\exists z) \neg P(z) \vee (\exists u)[\neg R(x,u) \vee (\forall z) \neg Q(u,z)]\} \\
 &\equiv (\exists x)(\forall y)\{(\exists z) \neg P(z) \vee (\exists u)[\neg R(x,u) \vee (\forall t) \neg Q(u,t)]\} \\
 &\stackrel{\text{extract quantifiers}}{\equiv} (\exists x)(\forall y)\{(\exists z) \neg P(z) \vee (\exists u)(\forall t)[\neg R(x,u) \vee \neg Q(u,t)]\}
 \end{aligned}$$

- Prenex forms:

$$U_2^{P_1} = \underbrace{(\exists x)(\forall y)(\exists z)(\exists u)(\forall t)}_{\text{prefix}} \underbrace{\{\neg P(z) \vee [\neg R(x,u) \vee \neg Q(u,t)]\}}_{\text{matrix}}$$

$$U_2^{P_2} = (\exists x)(\forall y)(\exists u)(\forall t)(\exists z)\{\neg P(z) \vee [\neg R(x,u) \vee \neg Q(u,t)]\}$$

- Skolem forms: $[x \leftarrow a, z \leftarrow f(y), u \leftarrow g(y)]$

$$U_2^{P_1} \text{ provides } U_2^{S_1} = (\forall y)(\forall t)\{\neg P(f(y)) \vee [\neg R(a, g(y)) \vee \neg Q(g(y), t)]\}$$

$$[x \leftarrow a, u \leftarrow g(y), z \leftarrow f(y)]$$

$$U_2^{P_2} \text{ provides } U_2^{S_2} = (\forall y)(\forall t)\{\neg P(g(y, t)) \vee [\neg R(a, f(y)) \vee \neg Q(f(y), t)]\}$$

- Clausal forms:

$$U_2^{C_1} = \neg P(f(y)) \vee [\neg R(a, g(y))$$

$$U_2^{C_2} =$$

- 2.2. Are the literals from the following pair unifiable? if yes, find their most general unifier.

$$x, y, z \in \text{Var}, \quad a, b \in \text{Const.}, \quad f, g \in F_1, \quad h \in F_2, \quad P \in P_3$$

$$\star l_1 = P(a, x, f(g(y))) \quad , \quad l_2 = P(y, f(z), f(z))$$

$$\theta_i = \varepsilon$$

- iteration 1:

$$\lambda_i = [y \leftarrow a], \quad \theta = \theta \lambda = [y \leftarrow a]$$

$$\theta(l_1) = P(a, x, f(g(a))) \quad , \quad \theta(l_2) = P(a, f(z), f(z))$$

• iteration 2:

$$\lambda := [x \leftarrow f(z)] \quad \theta = \theta\lambda = [y \leftarrow a][x \leftarrow f(z)] = [y \leftarrow a, x \leftarrow f(z)]$$

$$\theta(l_1) = P(a, f(z), f(g(a)))$$

$$\theta(l_2) = P(a, f(z), f(z))$$

• iteration 3:

$$\lambda := [z \leftarrow g(a)]$$

$$\theta = \theta\lambda = [y \leftarrow a, x \leftarrow f(z)][z \leftarrow g(a)] = [y \leftarrow a, x \leftarrow f(z), z \leftarrow g(a)] = \\ = \text{mgu}(l_1, l_2)$$

$$\theta(l_1) = \theta(l_2) = P(a, f(g(a)), f(g(a)))$$

common syntactic forms of l_1 and l_2

$$\star l_1 = P(x, g(f(a)), f(b))$$

$$l_2 = P(f(y), z, z)$$

$$\theta := \varepsilon$$

• it 1:

$$\lambda := [x \leftarrow f(y)]$$

$$\theta := \theta\lambda = [x \leftarrow f(y)]$$

$$\theta(l_1) = P(f(y), g(f(a)), f(b))$$

$$\theta(l_2) = P(f(y), z, z)$$

• it 2:

$$\lambda := [z \leftarrow g(f(a))]$$

$$\theta := \theta\lambda = [x \leftarrow f(y)][z \leftarrow g(f(a))] = [x \leftarrow f(y), z \leftarrow g(f(a))]$$

$$\theta(l_1) = P($$

$$\theta(l_2) = P($$

The terms $f(b)$ and $g(f(a))$ are not unifiable because none of them is a variable. We cannot unify the arg. on the last pos., so l_1 and l_2 are not unifiable.

3.2. Prove the inconsistency using lock resolution:

$$S_2 = \{ \overbrace{P(x)}^{c_1} \vee \overbrace{\neg Q(x)}^{c_2}, \overbrace{TP(a)}^{c_3} \vee \overbrace{R(x)}^{c_4}, \overbrace{Q(x)}^{c_5}, \overbrace{W(z)}^{c_6}, \overbrace{\neg R(y) \vee \neg W(y)}^{c_7} \}$$

$$C_6 = \text{Res}_{\theta_1 = [z \leftarrow y]}^{\text{lock}}(C_1, C_3) = P(x)$$

$$C_7 = \text{Res}_{\theta_1 = [z \leftarrow y]}^{\text{lock}}(C_1, C_5) = \neg R(y), \quad \theta_1 = \text{mgu}(W(z), W(y)) = [z \leftarrow y]$$

$$C_8 = \text{Res}_{\theta_2 = [x \leftarrow y]}^{\text{lock}}(C_2, C_7) = \neg R(y), \quad \theta_2 = \text{mgu}(R(x), R(y)) = [x \leftarrow y]$$

$$C_9 = \text{Res}_{\theta_3 = [x \leftarrow a]}^{\text{lock}}(C_6, C_8) = \square, \quad \theta_3 = \text{mgu}(P(a), P(x)) = [x \leftarrow a]$$

$$S_2 \vdash_{\text{Res}}^{\text{lock}} \square, \text{ so } S_2 \text{ is inconsistent}$$

4.2. Using a refinement of predicate resolution, prove the semidistributivity:

$$\vdash \mathcal{U}_1 : (\exists x)(P(x) \wedge Q(x)) \rightarrow (\exists x)P(x) \wedge (\exists x)Q(x)$$

$$\not\vdash \mathcal{U}_2 : (\exists x)P(x) \wedge (\exists x)Q(x) \rightarrow \exists x(P(x) \wedge Q(x))$$

$$\textcircled{T} \vdash \mathcal{U}_1 \text{ iff } (\neg \mathcal{U}_1) \vdash_{\text{Res}} \square$$

$$\begin{aligned} \neg \mathcal{U}_1 &= \neg (\exists x)(P(x) \wedge Q(x)) \rightarrow (\exists x)P(x) \wedge (\exists x)Q(x) \\ &\stackrel{\text{replace } \neg(\exists)}{\equiv} (\exists x)(P(x) \wedge Q(x)) \wedge \neg [(\exists x)P(x) \wedge (\exists x)Q(x)] \end{aligned}$$

$$\equiv (\exists x)(P(x) \wedge Q(x)) \wedge ((\forall y)\neg P(y) \vee (\forall z)\neg Q(z))$$

$$\equiv (\exists x)(\forall y)(\forall z)(P(x) \wedge Q(x) \wedge (\neg P(y) \vee \neg Q(z)))$$

$$(\neg \mathcal{U}_1)^S = (\forall y)(\forall z)(P(a) \wedge Q(a) \wedge (\neg P(y) \vee \neg Q(z)))$$

$$[x \leftarrow a], \text{ a - skolem const.}$$

$$(\neg \mathcal{U}_1)^C = P(a) \wedge Q(a) \wedge (\neg P(y) \vee \neg Q(z))$$

$$S_1 = C_1 = P(a)$$

$$C_2 = Q(a)$$

$$C_3 = \neg P(y) \vee \neg Q(z)$$